CHARACTERISATIONS OF CONSTRUCTIBLE SHEAVES ON A QUASI-COMPACT QUASI-SEPARATED SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the equivalence, including the refinement of the partitions and the remark that the same proof works for abelian sheaves and modules over a Noetherian ring.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Constructible sheaf and finite locally constant sheaf. Same chapter, the definitions of a constructible sheaf and of a finite locally constant sheaf on the small etale site. Supplies the three conditions compared in the statement.
- Constructible topology on the underlying space of a quasi-compact quasi-separated scheme. Same chapter together with the chapters Topology and Properties, the fact that the constructible subsets of such a scheme form a quasi-compact boolean algebra, which turns local information into a finite partition. Supplies the passage from the local condition (2) to the partition condition (3).

Source reference. Original lemma 09YD: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11742–11742. Navigation pointer invoked at proof line 11742; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 09Y4: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11745–11745. Navigation pointer invoked at proof

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

line 11745; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 054C: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11753–11753. Navigation pointer invoked at proof line 11753; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original definition 03RW: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11672–11672. Navigation pointer invoked at proof line 11755; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let X be a quasi-compact and quasi-separated scheme. Let $\mathcal{F}$ be a sheaf of sets on $X_{étale}$. The following are equivalent*

1. *$\mathcal{F}$ is constructible,*
2. *there exists an open covering $X = \bigcup U_i$ such that $\mathcal{F}|_{U_i}$ is constructible, and*
3. *there exists a partition $X = \bigcup X_i$ by constructible locally closed subschemes such that $\mathcal{F}|_{X_i}$ is finite locally constant.*

A similar statement holds for abelian sheaves and sheaves of Λ-modules if Λ is Noetherian.

Proof. It is clear that (1) implies (2).

Assume (2). For every $x \in X$ we can find an i and an affine open neighbourhood $V_x \subset U_i$ of x. Hence we can find a finite affine open covering $X = \bigcup V_j$ such that for each j there exists a finite decomposition $V_j = \coprod V_{j,k}$ by locally closed constructible subsets such that $\mathcal{F}|_{V_{j,k}}$ is finite locally constant. By Topology, Lemma 09YD each $V_{j,k}$ is constructible as a subset of X. By Topology, Lemma 09Y4 we can find a finite stratification $X = \coprod X_l$ with constructible locally closed strata such that each $V_{j,k}$ is a union of X_l. Thus (3) holds.

Assume (3) holds. Let $U \subset X$ be an affine open. Then $U \cap X_i$ is a constructible locally closed subset of U (for example by Properties, Lemma 054C) and $U = \coprod U \cap X_i$ is a partition of U as in Definition 03RW. Thus (1) holds. □